

HAMRing away at the speed of MACH 2

We're at the cusp of a major technological change in hard drive technology and the folks over at Seagate tell us what made it possible.

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d How is MACH 2 and HAMR going to affect the industry?

Jeff Nygaard: When you have a lot of data, part of the problem that arises is that the data centres would want to keep the IOPS/TB range between 5-10. Right now as we go through the 12-14 TB transitions, some of our customers are already seeing instances where they can't stay within range without stranding (not using) some capacity on the drive. MACH 2 is going to allow data centres to grow larger and maintain that service level for their end-customers. If you're a large data centre, then you can either grow your business by growing more storage and having a low service level, or you can service that by having a bunch of more super expensive SSDs, or you can have the right tiering of the data so that you can maintain that service level with the customer.

Mike Troemel: In the last 40 years we've had a six-million fold increase in storage density. HAMR is the next big technology that's going to keep that storage density growing. We're excited about it because we see the gains are slowing down with

the current technology. With HAMR we get back to a faster rate of growth. The higher capacity drives help our customers grow, it helps them with their data centre infrastructure challenges and helps reduce the overall TCO (Total Cost of Ownership).

d Multiple actuators seemed like an obvious way to advance hard drive technology. What made MACH 2 possible now?

Mike Troemel: There are a lot of challenges on the engineering and design side to make dual actuator drives work. One of the big challenges is the firmware, the MACH 2 technology is a significant increase in complexity. Managing the data flow and managing the performance of the two actuators are quite difficult. One of the enabling factors has been the firmware technology and the electronics associated with it. Getting enough memory and enough capability to control those actuators were really the fundamental advancements.

d Since the onboard buffer is capable of handling higher write

IOPS, was it the need to increase read IOPS that necessitated MACH 2?

Mike Troemel: Really both actually, our customers have been seeing potential throttling in read and write operations of single actuator devices. We're talking in terms of both, sequential and random characteristics.



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Mike Troemel
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d I believe this is one of the rare instances where a throughput focused technology as well as an IOPS focused technology are being brought out at the same time. In terms of R&D, what's next? Are you going to be increasing aerial density or IOPS?

Mike Troemel: They're both super critical for a storage device. We want to keep growing the capacity to meet the demands of the growing data requirements. Driving aerial density is a given, it's the lifeblood of our industry. And performance is also critical. As the capacity grows, the performance also has to keep up. So driving both is going to be fundamental.

d When it comes to transistors, the capability to shrink transistors has slowed down and it's no longer following Moore's law. We're

seeing companies struggle with 10nm and 7nm processes. In a similar vein, are you facing issues where it's becoming difficult to shrink the individual bits on the hard drive medium?

Mike Troemel: The current technology i.e. perpendicular magnetic recording, has been slowing down which is why we want to get to this newer technology to speed that up. If you think about NAND, it moved from a 2D planar technology to a 3D technology, and that allowed them to accelerate the growth. I'd say that HAMR (Heat-Assisted Magnetic Recording) is more analogous to that situation.

Jeff Nygaard: It's all about making the bit size on the storage device smaller and smaller. And to do that we have to go to different materials, and when you do that you need different energy profiles for the laser so that they can move the bits in a controlled way. We think that HAMR is our aerial density growth mechanism. Likewise with dual actuators, we have ways in which we can go further. We can go quad-actuators or we can have multiple heads per surface. There are a lot of things which we need to formalise with our customers to see if it serves their needs. **[E]**